Community Needs Dashboard

for Santa Clara's Valley Verde

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INTRODUCTION

The project supports Valley Verde in addressing food insecurity across the Bay Area, particularly in Santa Clara County. Despite Santa Clara County's location in Silicon Valley, food insecurity remains a persistent crisis. According to the 2025 Silicon Valley Index, the top 10% of earners hold 75% of the region's collective wealth, leaving roughly 33% of households unable to earn enough to be self-sufficient [1]. This, combined with the prevalence of food deserts in the San Jose area, has led organizations like Valley Verde to help families by providing raised-bed gardens, gardening supplies, and educational workshops on nutrition and sustainable food systems [2]. Currently, Valley Verde is developing a survey to gather information about the community's food needs and demographics. However, they lack a centralized system to integrate survey data and monitor trends over time regarding food scarcity from the community.

When researching potential solutions, the team examined other organizations' dashboards, which make data more digestible through visualizations. For instance, the U.S. Department of Agriculture's WIC Breastfeeding Dashboard presents key statistics with interactive maps, bar charts, and tables [3]. Similarly, Pew Research Center provides interactive maps to explore survey responses by city, offering insights into local demographics and opinions [4]. While these dashboards effectively visualize complex datasets, they are limited to single data sources and do not track longitudinal trends or provide insights for specific communities. These examples highlight the usefulness of the dashboard while underscoring the need for a tool that integrates multiple data sources and supports real-time community insights.

After reviewing the field, the team decided to create a dashboard that consolidates community survey responses and visualizes food needs and demographic trends over time

through graphs. It will help the Valley Verde team understand trends and better support community members that have limited access to nutritious food.

Based on Valley Verde's feedback, the team wants the dashboard to be visually appealing and easy to navigate for non-technical users. The dashboard must be free of cost for maintenance. It should be able to show survey data over time (weekly/monthly, or over total time) and be updated regularly on a daily basis. The dashboard's main goal is to show trends in both food insecurity as well as participants' demographics, allowing the partner to identify specific vulnerable populations. The data must also be accessible by community partners and the public as well. The dashboard will be successful if it's able to visually update survey responses in real time and is user-friendly for new users on the Valley Verde website

The primary users of the dashboard are the team members at Valley Verde. They will use the dashboard to consolidate responses from the community and better cater programs and events to suit the community's needs. Additionally, non-profit organizations, government agencies and researchers can view the data to better understand food insecurity in Santa Clara County. By displaying trends of community food needs and survey demographics through graphs and charts and generating exportable reports, the dashboard is able to give Valley Verde staff and members of the public the ability to monitor community needs over time.

This report breaks down the design and implementation of the dashboard. Methodology will expand on the tools and software the team used to implement the frontend and backend of the dashboard. In results, the report presents the findings from the dashboard in terms of efficiency, usability and performance. It expands on the visualizations in the discussion section and mentions how their results relate back to their project goals. It concludes by including future improvements for this project.

METHODOLOGY

Alternatives Analysis / Justification for Decisions

To evaluate the best solution, the team analyzed several data visualization approaches against the core criteria: cost effectiveness, easy handoff and low maintenance, future integration into Valley Verde's Wix website, and an accessible user experience for non-technical staff and visitors

According to Table 1 (see Appendix A), the alternatives were scored across four key criteria: usability, visibility, efficiency, and aesthetics, weighted at 50%, 25%, and 10% respectively. Based on the analysis, the "Do Nothing" option scored the lowest with a total of 4 points, reflecting minimal usability, visibility, efficiency, and aesthetic value. The mobile application scored moderately, achieving a total of 16 points, with strong efficiency and aesthetics but slightly lower usability and visibility compared to the web-based option. The web application outperformed all other alternatives with a total score of 19 points, demonstrating the highest usability and visibility, strong efficiency and excellent aesthetics. This combination indicates that the web application best aligns with the team's priorities, offering an accessible, visually clear, and user-friendly solution that can be integrated seamlessly into Valley Verde' existing website.

Prototype Development

The two prototyping platforms that the team considered were Figma [5] and AI Studio [6]. To determine the most effective approach, the team evaluated options such as static mockups and interactive dashboard prototypes based on criteria including ease of use for non-technical audiences, the ability to simulate realistic data visualizations, and how easily the design could

translate into a functional web dashboard. Based on this evaluation, the team selected an interactive prototype developed in AI Studio [6]. This approach allowed the team to quickly iterate on layout and visualization concepts while demonstrating how survey data could appear in the final product. Sample responses were used to simulate how data would be displayed, allowing Valley Verde to review the design and provide feedback to ensure the dashboard aligned with their needs.

The backend prototype was created in Postman [7] to finalize the response structure. The team used a variety of project management and software development platforms (see Appendix D for a complete list of tools and technical documentation as well as analysis for choosing these final tools) to implement the dashboard system. The Google Forms [8] were used to conduct the survey and collect data. React [9] and AI Studio [6] were used to build the code. The backend utilized FastAPI [10] and Google Sheet APIs [11] to process and manage the data. For easier collaboration, the team consolidated code in a GitHub [12] repository and shared resources via Google Drive [13]. Tasks and timelines were organized using a Gantt chart on Monday.com [14]. Project development tasks were divided among the five team members to optimize efficiency (see Appendix B for a detailed breakdown of individual roles).

With a $100 budget, Valley Verde prioritizes keeping costs low, so the team used existing tools and avoided AI integration. Within the ten-week timeline, the project focused on creating a dashboard for a survey with a limited set of multiple-choice questions provided by Valley Verde.

The final dashboard visualizes community food needs and demographics, ensuring accessibility for all Santa Clara residents. The fully functional web application, deployed on Vercel and Render, provides interactive access to dynamic visualizations.

Testing Protocols and Design of Experiment (DOE)

To ensure the dashboard meets the community partner's needs, the team designed a series of experiments evaluating the system against four main project goals. Each goal is paired with a specific testing method and clear measure of success.

Testing the Data Pipeline

To verify that the dashboard is able to accurately translate community survey data, the team tested the end-to-end data flow. First, multiple sets of sample data were submitted through the Google Form [8] to simulate real user input. These submissions included varied responses to test different data scenarios and edge cases. After each submission, the dashboard was monitored to verify that the data was updated in real time. The success metric is being 100% accurate in automatically pulling and rendering the submitted mock data.

Cost Analysis

To ensure the project's costs remain within the budget constraint, the team audited all materials and tools used. The success metric is showing a cost basis of below $100 for total product development as well as minimal operational/maintenance costs.

Testing User Interface and Data Clarity

The team conducted a usability test with 10 participants to evaluate the dashboard's accessibility for non-technical users. Participants accessed the live Vercel deployment URL and followed testing instructions(see Appendix I). They navigated the dashboard pages and completed a Google Form [8] feedback survey, rating their experience on a 5-point scale (1=Very difficult/unclear, 5=Very easy/clear) using targeted quantitative questions, including evaluating the clarity of charts, ease of interpretation, navigation of the dashboard layout, and how quickly users could locate the needed information.

The survey also includes open-ended questions to gather design feedback, asking users what features they would like added and how the dashboard could be improved. Success was defined as an average rating of 4.0 or higher across all quantitative questions, indicating high level of accessibility and intuitive design.

RESULTS

Before presenting the detailed findings, it is important to understand how the final prototype functions. The dashboard is a fully functional, interactive web application that collects survey data, processes it, and presents it in an accessible visual format. By translating raw Google Form [8] responses into meaningful metrics, such as food insecurity levels and affordability of specific foods, the dashboard provides clear, actionable insights. The following section presents the results of the project, including the visual outputs generated by the dashboard and key observations gathered during prototype testing.

Final Prototype

The final prototype is a fully functional, interactive web application deployed via Vercel and Render. The outcomes of the project are evaluated based on the visual outputs of the dashboard and the performance metrics gathered during testing. The final prototype successfully resolves Valley Verde's need for a data analysis tool by creating an automated dashboard that translates raw Google Form [8] survey responses into dynamic, easily digestible visualizations.

The dashboard works by collecting survey responses through a Google Form [8], which are automatically stored in an associated Google Sheet. Each response is accessed by the backend through the Google Sheets API [11]. The backend, implemented using FastAPI [10], processes the data to generate meaningful metrics and insights, such as levels of concern about food insecurity and specific foods that respondents cannot afford. These processed metrics are

set on the front-end, where they are displayed through interactive visualizations, including pie charts, bar graphs, and other dashboard elements. The fully functional web application was deployed on Vercel and Render, allowing interactive access to all features and dynamic visualizations

The dashboard interface is divided into four sections to organize information and increase accessibility.

Section 1: Home Page. The Home Page is the initial screen where it provides a quick summary about the dashboard, how to use the dashboard, and a link to the Google Form [8] survey. (see Appendix E)

Section 2: Community Food Needs. The Community Food Needs contains metrics related to food insecurity. (see Appendix F)

Section 3: Demographics and Geography. The Demographics and Geography page displays demographic data of the survey respondents. (see Appendix G)

Section 4: Timeline. The Timeline page displays the overall timeline of how community food security concerns have changed over time. (see Appendix H)

System Testing and Evaluation Metrics

To verify that the dashboard meets all project criteria and community partner needs, the prototype was rigorously evaluated against both technical and user-based success metrics.

Data Pipeline Validation

System testing confirms that the prototype operates with 100% data transmission accuracy. Simulated survey entries submitted through the embedded Google Form [8] successfully log into the Google Sheet, route through the FastAPI [10] backend, and render on

the React [9] frontend within moments. The system requires zero manual data manipulation from the user, successfully automating the organization's data analysis.

Cost Analysis

Deployment audits confirm the system successfully operates with zero-cost maintenance. The application runs entirely on cost-free infrastructure, leveraging Vercel for frontend hosting and Google Workspace APIs for backend database management. This architecture ensures Valley Verde incurs no costs to use and maintain the dashboard.

Usability Data

To test the dashboard's user experience, 10 participants tested the deployed website and provided feedback using a 1-to-5 scale. The scores show that the design is intuitive, with the dashboard exceeding the 4.0 average success goal across all tested areas. (see Appendix J)

Visual Effectiveness. Chart clarity and visual appeal receive the highest average scores (4.7 out of 5), with 70% of participants giving the system a perfect 5. This proves the dashboard successfully turns complex numbers into clear pictures.

Helpfulness and Workflow. The system averages a 4.6 out of 5 for overall usefulness and how well it supports the user's workflow. 100% of users rate these categories as a 4 or 5, proving the dashboard effectively visitors quickly understand community food need data.

Finding Information. The speed and ease of finding specific information both average a 4.5 out of 5. The ease of moving around the dashboard and the ease of understanding the graphs both average 4.4 out of 5.

User Experience

While the testing scores are highly positive, user feedback reveals a few specific areas that can be improved in the future. (see Appendix K)

Mobile Layout Issues. Users note that testing on a mobile phone shows a layout issue where the “Need Trend Guide” ranking gets pushed to the bottom of the screen, instead of staying on the right side where it belongs.

Button Clarity. Users suggest spelling out button functions instead of just using an icon so visitors know exactly what the button does.

More Detailed Filters. Users express a strong desire for more ways to sort the data. They specifically asked for filters based on background (race, gender, age) and location (city, suburban, rural) to get a better, more detailed understanding of local food needs.

DISCUSSION

Interpretation of Results

The testing results clearly demonstrate that the final dashboard successfully meets all core project objectives. By achieving high usability scores (averaging above 4.4 out of 5 in all categories), the dashboard proves to be highly accessible for non-technical users. More importantly, the 100% success rate of the automated data pipeline confirms that the system resolves Valley Verde’s primary need for a data analysis tool. The data flows directly from the Google Form [8] to the visual dashboard, saving the organization valuable time and allowing them to focus directly on community outreach. Furthermore, by utilizing free-tier hosting and database tools, the project successfully delivers a solution with zero maintenance costs, following the organization's financial constraints.

Interpretation of Findings

While the dashboard performs well in its core functions, user testing revealed a few limitations. Similar shades of green on the "Area change over time" graph make overlapping trends hard to distinguish, and the mobile layout pushes the “Need Trend Guide” to the bottom of

the screen. These issues indicate that, although highly functional on desktop computers, the dashboard's color palette and responsive design need minor adjustments to improve accessibility on all devices.

Recommendations for Future Development

Based on the qualitative feedback collected during testing, future iterations of the dashboard should focus on expanding the data filtering capabilities. Adding specific demographic filters (such as race, gender, and age) and geographic filters (such as urban, suburban, and rural) would allow Valley Verde to gain a much deeper understanding of exactly who is experiencing food insecurity and where they are located. Additionally, minor user interface updates, such as explicitly spelling out "Print Report" and ensuring all bar graph labels are visible without clicking, will further reduce confusion and improve the overall user experience.

Connect to Goals

The project's direction went from an app to a web application because of the time it took to work on this project, and how a web application would be easier to integrate into the Valley Verde website. Success was measured by the usability of the dashboard , being able to integrate the front end and back end, and incorporating the dashboard to the Valley Verde website.

In the past weeks, the team narrowed down the scope of the project, made clear goals, created a working prototype and have a working frontend and backend code. The success of the project was measured by creating a working model that reflected the advice from their community partners. Valley Verde played a big role in helping better define the problem statement, and giving questions in the Google Form [8] used for data collection. Their feedback on the types of graphs shaped the visualization of the data on the dashboard. Based on the class

feedback given during the presentation, the final group presentation has a simple breakdown of the technical aspects of the project. Clearly explaining the frontend and backend logic and how it relates to the bigger picture.

CONCLUSION

The initial problem was that the staff at Valley Verde needed a program to develop community survey tools and consolidate information on food insecurity in the Bay Area in real time. The project addresses the issue by creating a dashboard that consolidates information from a Google Form [8] on food insecurity and displays trends through graphs and charts. The automated interface allows Valley Verde to quickly visualize trends and extract actionable insights, enabling their team to respond more effectively to the community's needs.

Throughout the quarter, the team adapted to several technical challenges and adjusted team norms to ensure success. A reflection on the design process and individual lessons learned is available in Appendix C. For future development, the dashboard could be expanded to incorporate qualitative data from community member interviews. By transcribing conversations and implementing keyword extraction, recurring themes could be visualized alongside survey trends. Additionally, transferring the project to Valley Verde's GitHub [12] repository will allow future student teams or developers to maintain, expand, and refine the tool over time.

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APPENDIX

Appendix A: Alternatives Analysis

	Criteria	C1 Usability (50%)	C2 Visibility (25%)	C3 Efficiency (15%)	C4 Aesthetic (10%)	Total 100
Alternatives	Do Nothing	1	1	1	1	4
	Web application	5	5	4	5	19
	App	3	4	5	4	16

Appendix B: Team Roles and Responsibilities

The team split roles based on our skills and availability. Conner was the group's community liaison and acted as the primary point of contact for the community partner. He also worked closely with Backend and Frontend Lead in building the dashboard. Lily was the team's project manager. She helped facilitate meetings, take meeting notes and tracked deadlines and progress. Irene was the backend lead for the team. She oversaw the backend development, managed data processing workflows, and helped design the data analysis portion of this project. Shibo was the team's front-end developer. He used React [9] and D3.js to establish stable connections with the backend and create intuitive UI/UX design for data visualization.

Appendix C: Team Reflections

Reflecting on the quarter, the team's success was rooted in establishing team norms. The team prioritized transparent communication and regular check-ins, which helped the team navigate the technical challenges of linking the form data to the visualization. One of the biggest successes was the team's ability to pivot when the team realized a certain feature or tactic

wouldn't be the most usable or efficient in communicating the 'why' behind the data. The team learned to view the work through the eyes of the Valley Verde staff who would be using the dashboard.

Lessons Learned:

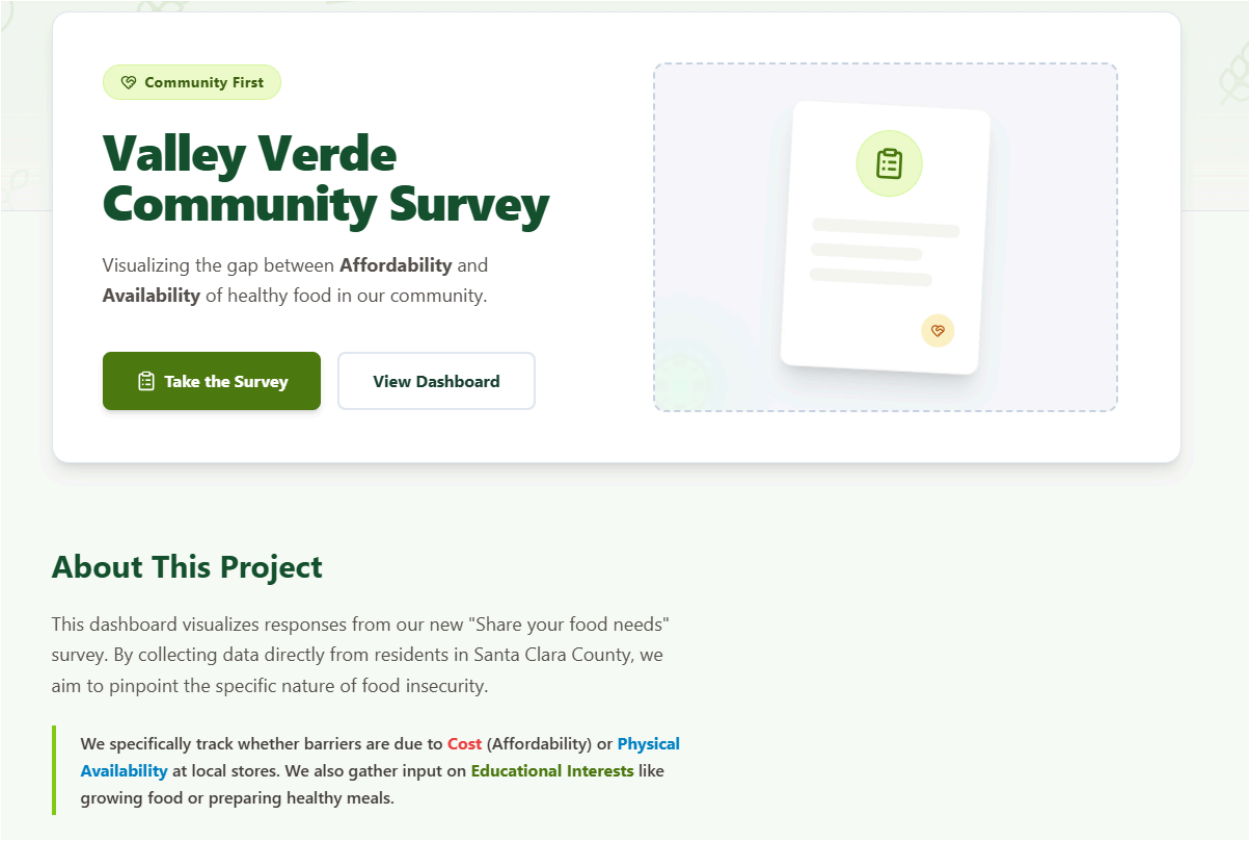
Team	• Managing and limiting the scope of their project to fit the time and budget constraints • Data ethics and the social impact • Considering handoff as well as functionality • Communicate technical concepts in lay language
Conner	• Learned to pivot design and implementation decisions based on feedback, ensuring features are practical and user-friendly • Gained awareness of how the way data is visualized affects interpretation as well as the importance of responsibly presenting community information. • Introduced to design and prototyping tools like Figma [5] and AI Studio [6].
Lily	• Gained experience managing deadlines and responsibilities with project management tools like Monday [14] • Learned to shift strategies based on feedback and project scope. Pivoted from a web application to a dashboard that can be embedded into the partner's existing Wix website.
Irene	• Clear and consistent communication is essential to team's success • Improved programming skills through hands-on collaboration • Gained confidence in solving technical problems
Shibo	• Developed an understanding of the essential process required to execute a successful project through effective teamwork. • Gained valuable experience in project management, including the use of Gantt charts for structured planning and coordination.

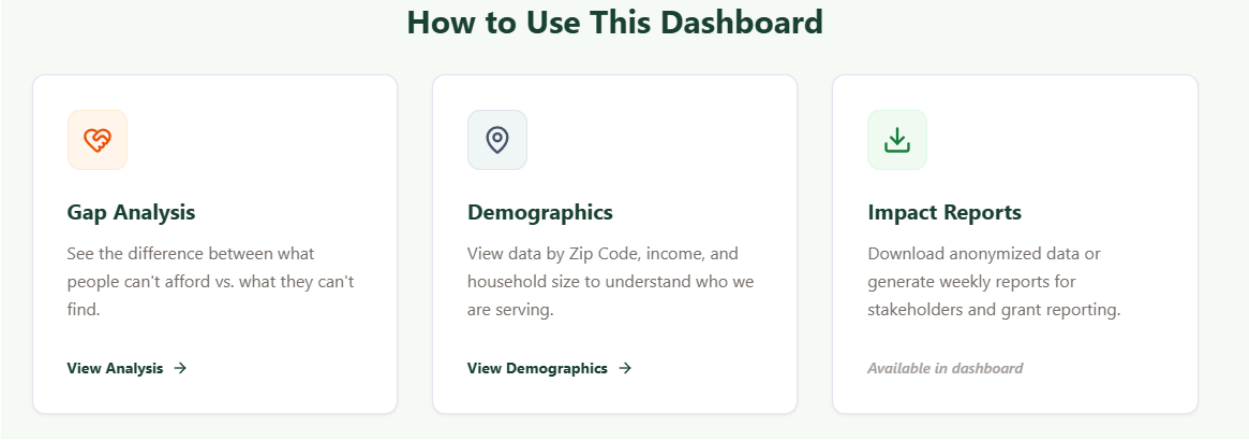
	• Strengthened the ability to design more professional, user-centered, and visually refined UI/UX interfaces.
Anushri	• Learned about data visualization • Learned to translate technical ideas into a simpler system • Importance of Food insecurity programs in the Santa Clara Community

Appendix D: Technical Tools and Platforms

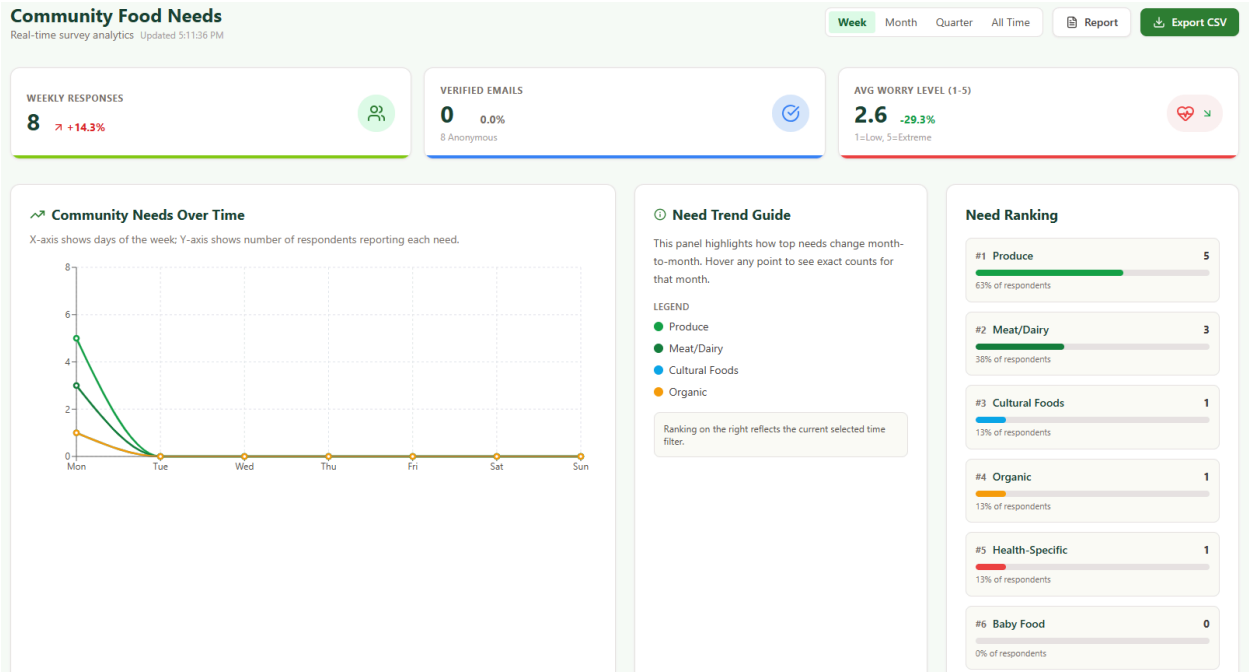
Tool/Platform	**Purpose**
GitHub [12]	Source code management, collaboration, and version tracking. Optimal for team collaboration since everyone can have access to the repository and commit changes individually.
Google Forms [8]	Where the team is conducting the survey for the project to get sample data. Google Forms was chosen for its ease of use and seamless integration with Google Sheets.
Google Drive [13]	Planning and documentation, allowing all team members to have easy access and share project files in a centralized and organized manner.
AI Studio [6]	Designing and prototyping the user interface for the dashboard, enabling rapid iterations and visualization for the product's layout and design concepts.
Monday.com [14]	Task assignment, timeline tracking, and team coordination, helping the team organize workflows and monitor project progress effectively.
FastAPI(Python) [10]	Implementing backend logic and API endpoints, providing a lightweight and efficient framework for handling data processing and communication.
React [9]	Building the user-facing interface with flexibility and ability to create dynamic, interactive user interfaces.
Google Sheet API [11]	Used to read data from the Google Sheet with low-cost, low-maintenance data management.

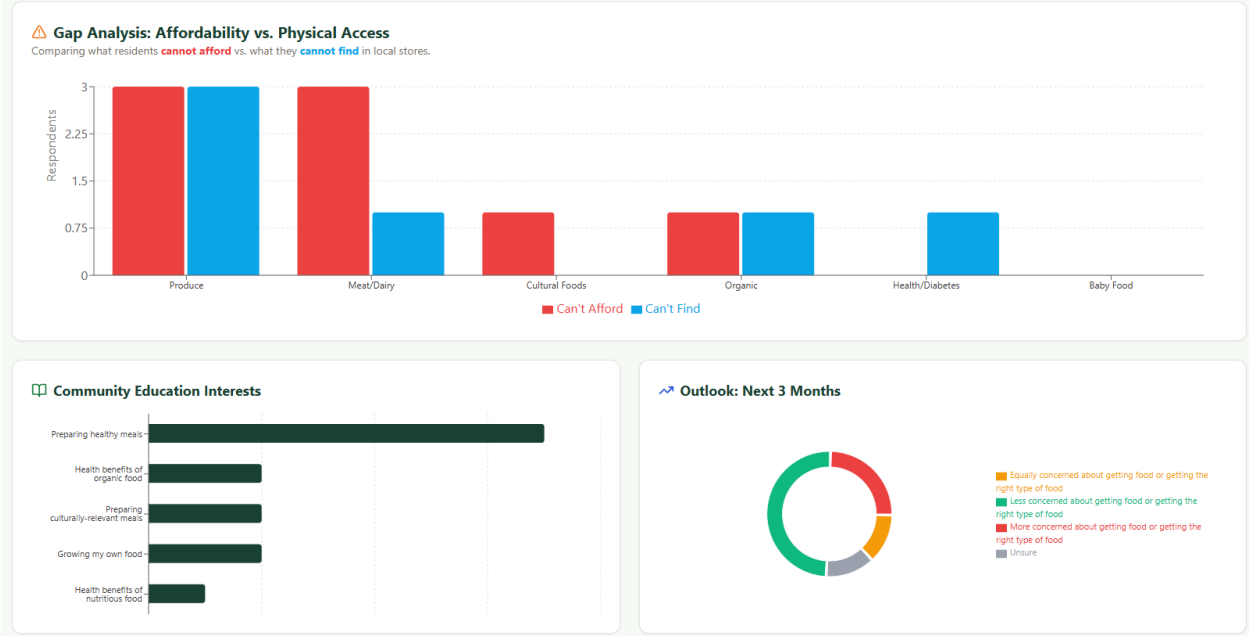
Appendix E: Home Page



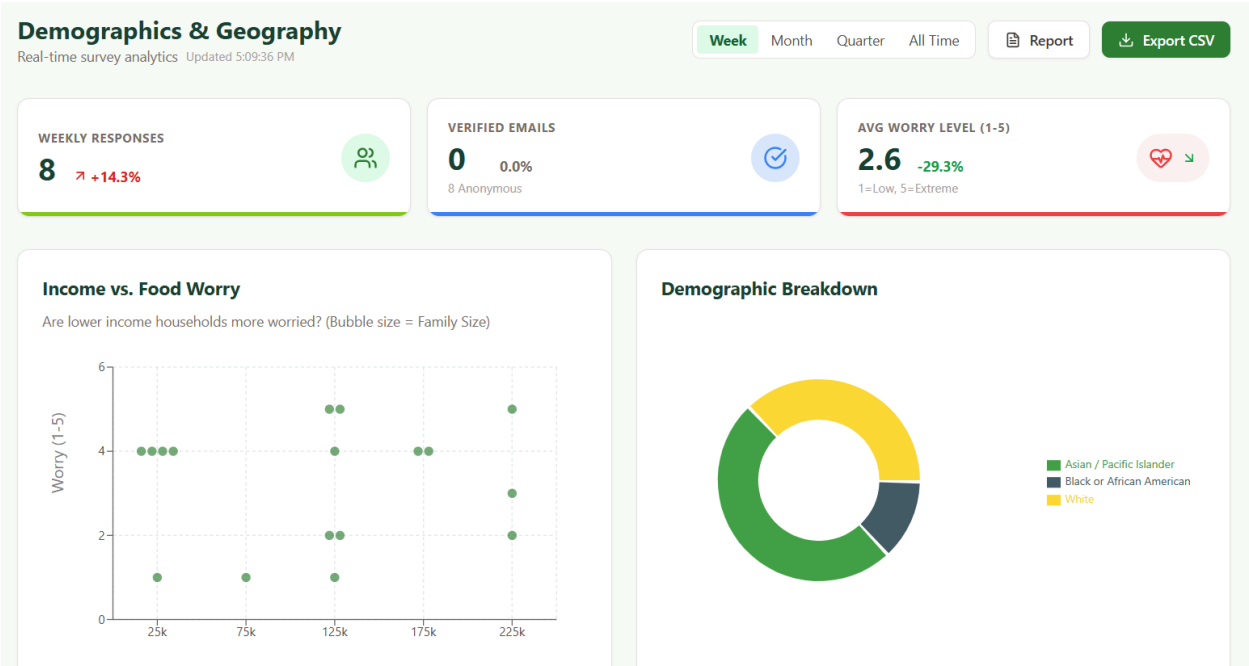


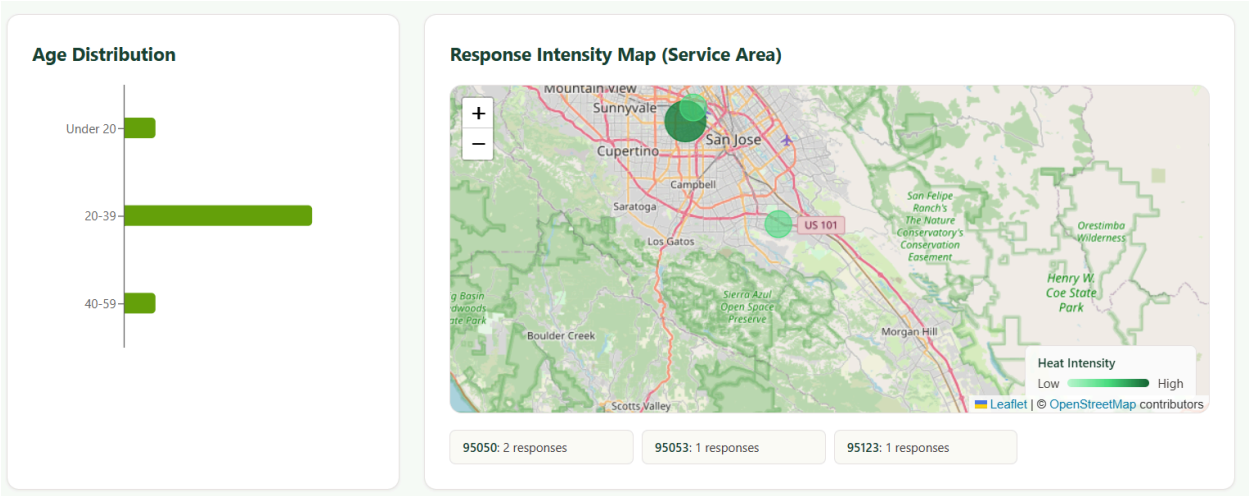
Appendix F: Community Needs Page





Appendix G: Demographics Page


Demographics & Geography
Real-time survey analytics Updated 5:09:36 PM
Week
Month
Quarter
All Time
Report
Export CSV
WEEKLY RESPONSES
8 +14.3%
VERIFIED EMAILS
0 0.0%
8 Anonymous
AVG WORRY LEVEL (1-5)
2.6 -29.3%
1=Low, 5=Extreme
Income vs. Food Worry
Are lower income households more worried? (Bubble size = Family Size)
Worry (1-5)
0
2
4
6
25k
75k
125k
175k
225k
Demographic Breakdown
Asian / Pacific Islander
Black or African American
White


Age Distribution
Under 20
20-39
40-59
Response Intensity Map (Service Area)
Mountain View
Sunnyvale
San Jose
Cupertino
Campbell
Saratoga
Los Gatos
US 101
San Felipe Ranch's The Nature Conservatory's Conservation Easement
Orestimba Wilderness
Henry W. Coe State Park
Sierra Azul Open Space Preserve
Boulder Creek
Morgan Hill
Scotts Valley
Heat Intensity
Low
High
Leaflet | © OpenStreetMap contributors
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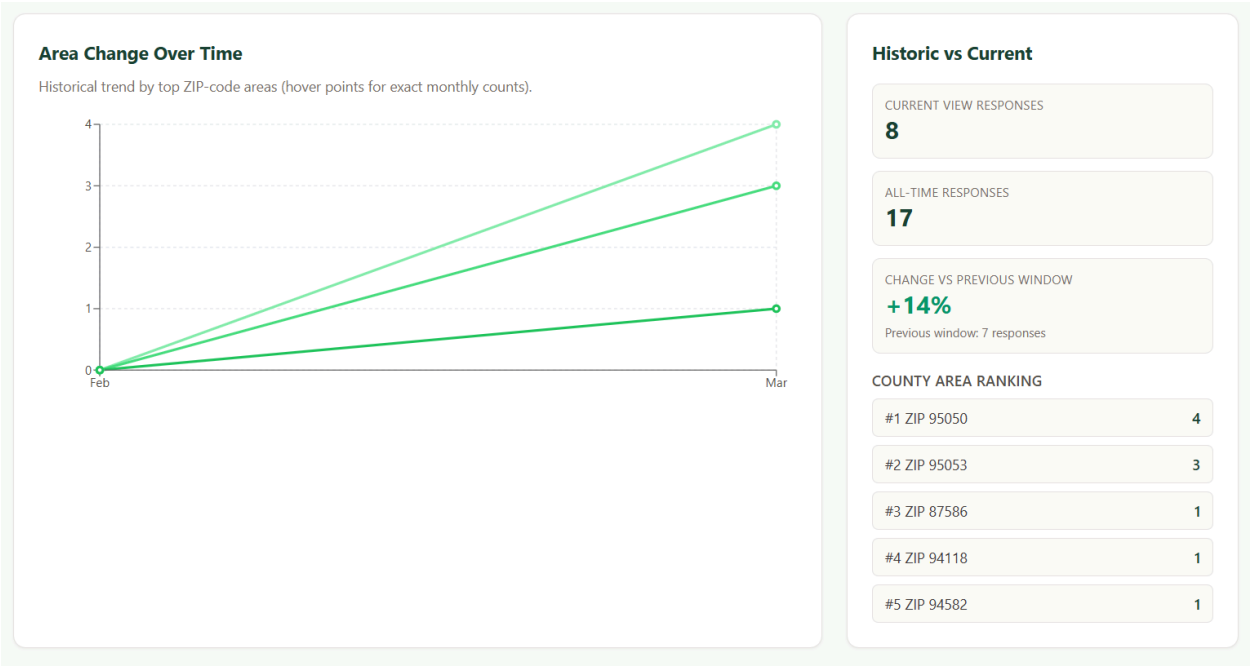
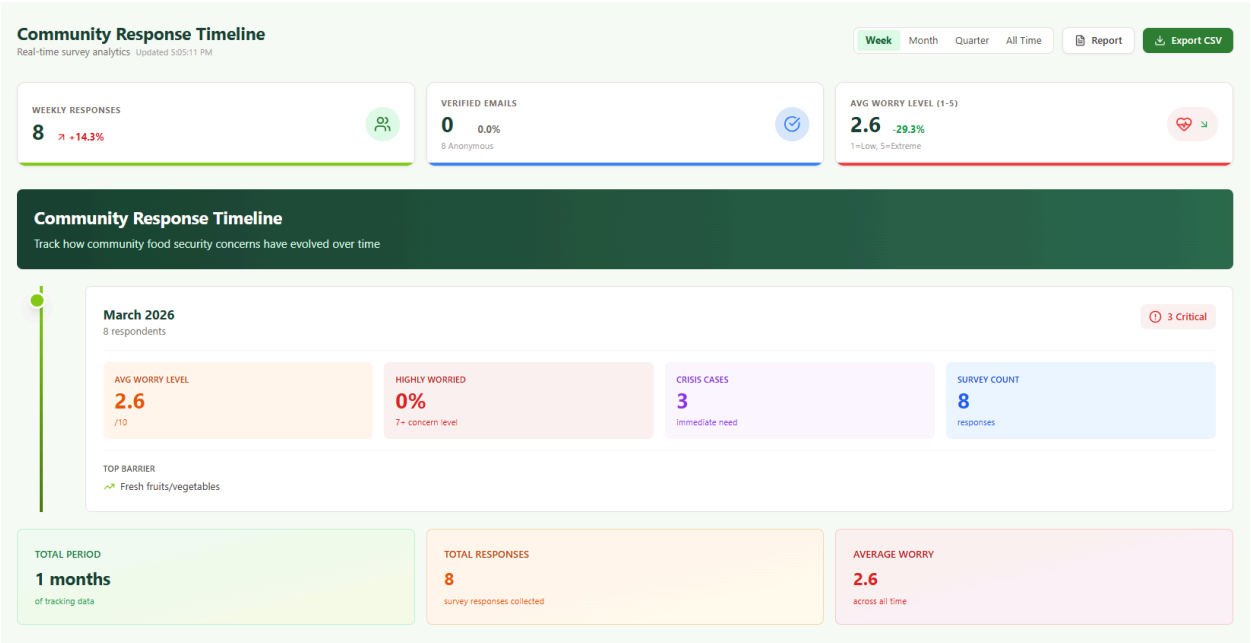
95053: 1 responses
95123: 1 responses



Appendix H: Community Timeline Page